YEAR 7 MATHS · CHAPTER 8 · MIND MAP

Statistics and probability

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1 Collecting and classifying data

Categorical data is groups. Numerical data is numbers.

Categorical: favourite sport

Numerical: discrete (counts), continuous (measurements)

Primary: you collect it. Secondary: someone else did.

2 Summarising data

Mean: add and divide. Median: middle in order. Mode: most common. Range: max − min.

2, 3, 3, 6, 11: mean $\frac{25}{5} = 5$

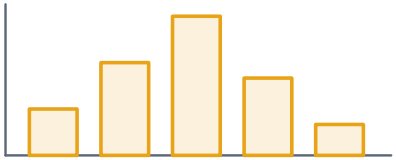
Median 3, mode 3, range 9

3 Column graphs and dot plots

Give it a title, label both axes and use an equal scale.

Column graphs: leave gaps between columns

Dot plots: one dot for each result



4 Line graphs

Show change over time, with time on the horizontal axis.

Look for rising, falling and flat parts



5 Stem-and-leaf plots

The stem is the tens, the leaf is the units. Include a key.

1 | 2 5 8 and 2 | 0 3 (key 1 | 2 = 12)

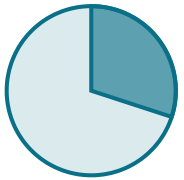
Data 12, 15, 18, 20, 23: median **18**

6 Sector and divided bar graphs

Each part of the circle or bar shows a share of the whole.

A quarter of a circle is $90°$

30% of a circle: $0.3 \times 360° = 108°$



30% of the circle
30% × 360° = 108°

7 Describing probability

Probability runs from 0 (impossible) to 1 (certain).

Unlikely, even chance $\left(\frac{1}{2}\right)$, likely

8 Theoretical probability

$P = \frac{\text{favourable}}{\text{total}}$ when outcomes are equally likely.

Fair die: $P(3) = \frac{1}{6}$, $P(\text{even}) = \frac{1}{2}$

3 red and 2 blue: $P(\text{red}) = \frac{3}{5}$

$P(\text{not } A) = 1 - P(A)$

9 Experimental probability

Relative frequency $= \frac{\text{times it happened}}{\text{number of trials}}$

12 heads in 40 tosses: $\frac{12}{40} = 0.3$. The theoretical probability is 0.5.

More trials usually get closer to the theoretical probability.